Consumption of Brussels sprouts protects peripheral human lymphocytes against 2-amino-1-methyl-6-phenylimidazo[4,5-b]pyridine (PhIP) and oxidative ...

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To find out if the cancer protective effects of Brussels sprouts seen in epidemiological studies are due to protection against DNA-damage, an intervention trial was conducted in which the impact of vegetable consumption on DNA-stability was monitored in lymphocytes with the comet assay. After consumption of the sprouts (300 g/p/d, n = 8), a reduction of DNA-migration (97%) induced by the heterocyclic aromatic amine 2-amino-1-methyl-6-phenyl-imidazo-[4,5-b]pyridine (PhIP) was observed whereas no effect was seen with 3-amino-1-methyl-5H-pyrido[4,3-b]-indole (Trp-P-2). This effect protection may be due to inhibition of sulfotransferase 1A1, which plays a key role in the activation of PhIP. In addition, a decrease of the endogenous formation of oxidized bases was observed and DNA-damage caused by hydrogen peroxide was significantly (39%) lower after the intervention. These effects could not be explained by induction of antioxidant enzymes glutathione peroxidase and superoxide dismutase, but in vitro experiments indicate that sprouts contain compounds, which act as direct scavengers of reactive oxygen species. Serum vitamin C levels were increased by 37% after sprout consumption but no correlations were seen between prevention of DNA-damage and individual alterations of the vitamin levels. Our study shows for the first time that sprout consumption leads to inhibition of sulfotransferases in humans and to protection against PhIP and oxidative DNA-damage.